TimeTensors Experiment 5: Linear Models

1 Theoretical overview and goal

Strong linear controls reveal whether gains attributed to complex architectures come from nonlinear representation or simply from autoregressive structure. Persistence repeats the latest observation. A learned linear head maps the whole look-back to the horizon. Periodic linear models restrict each horizon step to matching seasonal phases. Scikit-learn regression provides a deterministic full-window fit independent of the PyTorch sampling loop.

Experimental goal. Establish transparent capacity and optimization baselines, test the value of explicit periodic structure, and detect implementation gaps between sampled and deterministic linear fitting.

2 Implementation details

The comparison contains persistence, learned linear, periodic linear, and scikit-learn multi-output linear regression. The scikit-learn path unrolls all sampler-accessible training windows deterministically and applies the same normalization/inverse-normalization contract as PyTorch models. Periodic heads receive only look-back positions with the appropriate phase.

The same datasets and (L, H) settings as the methodology study are used. Learned and deterministic linear models are compared under identity, global standard, and instance normalization, with a separate coefficient plot for each fitted configuration. Learned models use the common learning rate, batch size, optimizer-step budget, evaluation frequency, and three seeds; deterministic methods naturally have zero seed variance unless data selection is randomized.

Launcher: `06_linear_models.slurm`

Because this is a comparison between model families rather than DLinear and PatchTST variants, all methods share one table. Report test MSE, parameter or fit type, seed mean $\pm$ standard deviation, and per-row multiplier.

3 Interpretation

Compare learned and deterministic linear fits to identify optimization effects. Compare periodic and unrestricted heads to assess the usefulness of the assumed season. A nonlinear model should be judged against the best linear control, not persistence alone.